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B-4 Country Home, Bhuhera Road, Barabanki, Nawabganj, Uttar Pradesh - 225003

INTERNSHIP REPORT ON

Robotics and Automation

Projects Undertaken

Smart Home Lighting System

Conveyor Belt with Sensors

Automated Packaging System

Submitted by:

SESHANAND S

Student ID:CA/DF1/54581

College: Sathyabama Institute of Science and
Technology, Chennai

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ABSTRACT

The internship in Robotics and Automation provided practical exposure to automation technologies, sensor interfacing, and circuit designing. During the internship, various automation-based systems were designed and simulated to understand the working principles of smart and industrial automation applications.

The major projects completed during the internship include a Smart Home Lighting System, Conveyor Belt with Sensors, and an Automated Packaging System. These systems were developed using automation components such as sensors, microcontrollers, relays, motors, and control circuits. The projects focused on automatic operation, energy efficiency, object detection, and process control.

The Smart Home Lighting System was designed to automatically control lighting based on motion detection and ambient light conditions. The Conveyor Belt System used sensors to detect objects and automate conveyor movement. The Automated Packaging System demonstrated automatic product handling and packaging operations using sensor-based control mechanisms.

The circuits and simulations were developed using [Tinkercad](#), which helped in understanding real-time circuit behavior and automation logic. Through this internship, knowledge was gained in robotics, automation systems, embedded control, sensor integration, and industrial applications. The internship also improved technical skills, problem-solving abilities, and understanding of modern automation technologies.

INTRODUCTION

Robotics and Automation play a major role in modern industries, smart homes, and manufacturing systems by improving efficiency, accuracy, and productivity while reducing human effort. Automation systems use sensors, controllers, and actuators to perform operations automatically with minimal human intervention. These technologies are widely used in industrial production lines, intelligent home systems, packaging industries, and smart monitoring applications.

During the internship in Robotics and Automation, various automation-based circuits and systems were designed and simulated to understand the practical implementation of automated control systems. The internship focused on circuit designing, sensor interfacing, automation logic, and system simulation using modern tools and components.

Three major projects were completed as part of the internship:

1. **Smart Home Lighting System** – designed to automatically control room lighting based on motion detection and ambient light conditions for energy-efficient operation.
2. **Conveyor Belt with Sensors** – developed to automate material movement using sensors for object detection and conveyor control, similar to systems used in industrial production units.
3. **Automated Packaging System** – designed to demonstrate automatic product handling and packaging operations using sensors and motor control mechanisms.

These projects provided practical exposure to robotics and automation concepts such as sensor integration, embedded control systems, automatic switching, industrial automation techniques, and real-time process monitoring. The circuits and simulations were created using [Tinkercad](#), which enabled efficient testing and analysis of the designed systems.

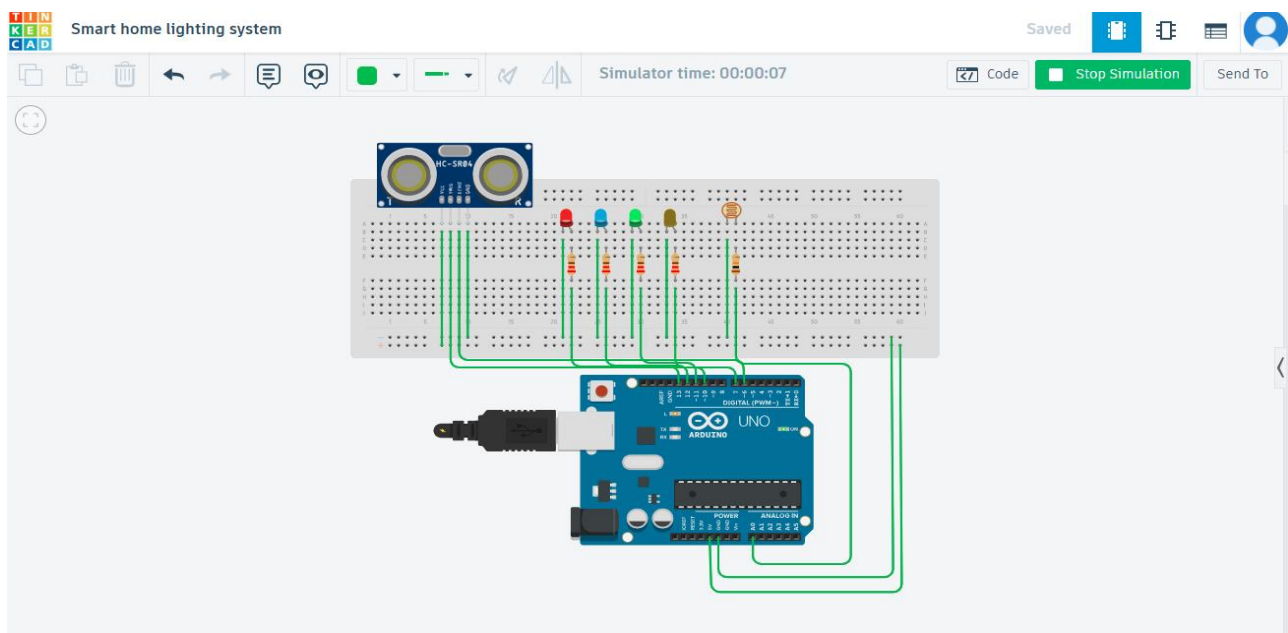
The internship helped in understanding the importance of automation in modern engineering applications and improved technical knowledge in circuit designing, troubleshooting, and automation system development.

Project 1 – Smart Home Lighting System

Description

This project demonstrates an automated smart lighting system using an Arduino UNO, ultrasonic sensor, LDR sensor, and LEDs. The system automatically controls lighting based on object detection and ambient light intensity.

Tinkercad link: <https://www.tinkercad.com/things/k2MUzgRsGU1-smart-home-lighting-system>



Components Used

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- LDR Sensor
- LEDs
- Resistors
- Breadboard
- Connecting Wires

Working Principle

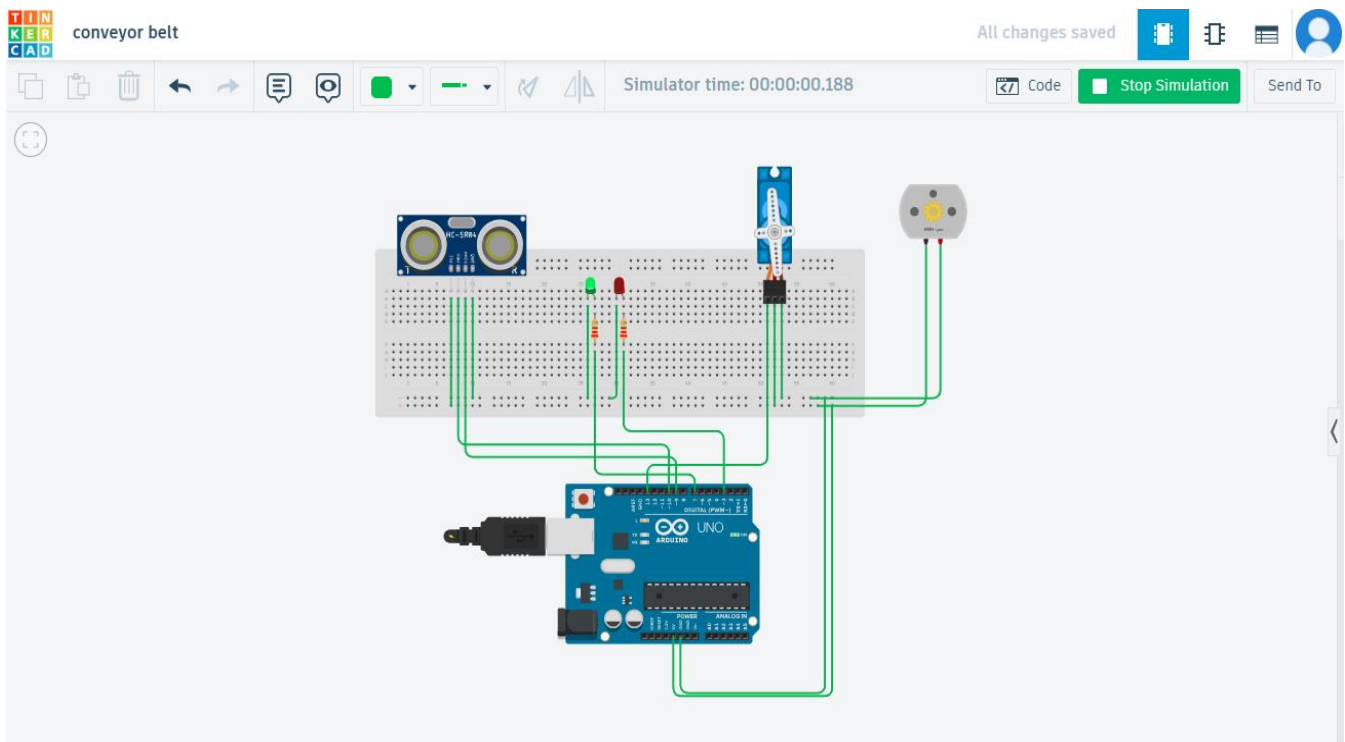
The ultrasonic sensor detects the presence of a person or object, while the LDR sensor measures ambient light intensity. When motion is detected in a dark environment, the Arduino activates the LEDs automatically. The system improves energy efficiency by reducing unnecessary power consumption.

Project 2 – Conveyor Belt with Sensors

Description

This project represents an automated conveyor belt system using sensors and motor control mechanisms. The system detects objects and controls conveyor movement automatically.

Tinkercad link: <https://www.tinkercad.com/things/jBnkbFxeipz-conveyor-belt->



Components Used

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Servo Motor
- DC Motor
- LEDs
- Resistors
- Breadboard

Working Principle

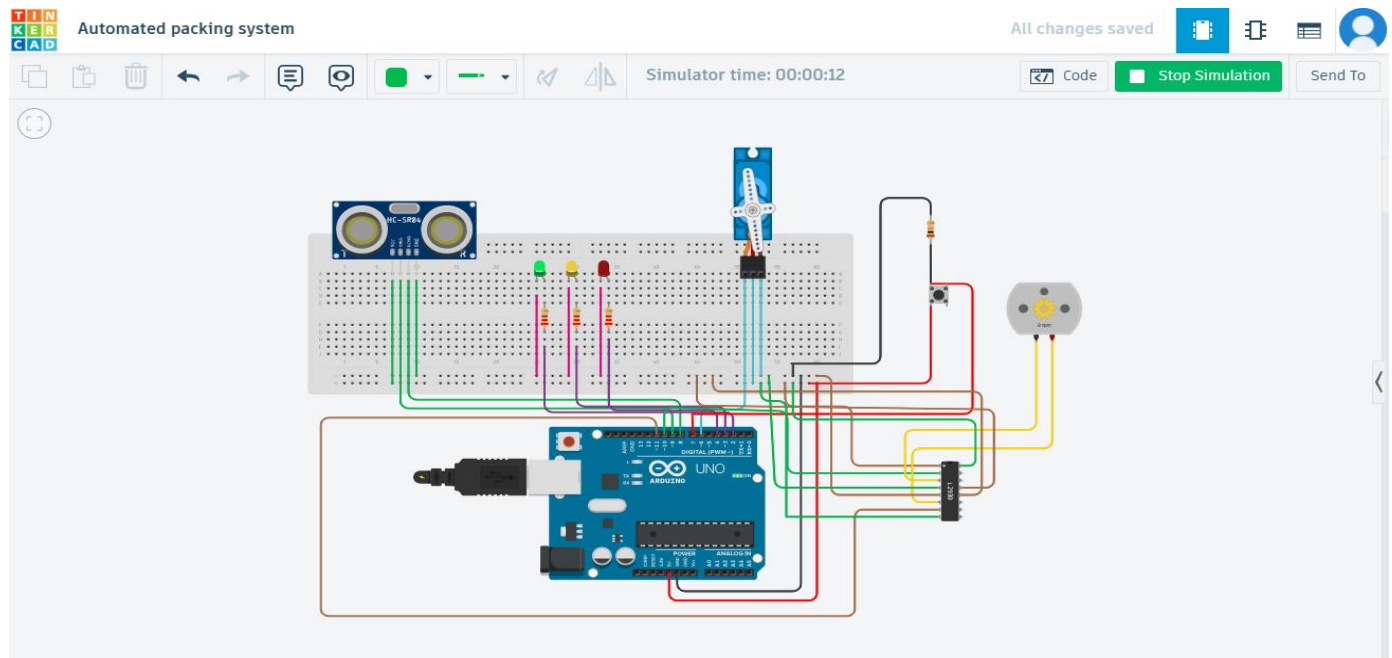
The ultrasonic sensor detects incoming objects on the conveyor path. Based on sensor input, the Arduino controls the servo motor and conveyor mechanism. Indicator LEDs display system status such as object detection and movement operation.

Project 3 – Automated Packaging System

Description

This project demonstrates an automated packaging system using Arduino, sensors, servo motor control, and motor driver circuits for automated product handling and packaging operations.

Tinkercad link: <https://www.tinkercad.com/things/kVXv7PwvhVN-automated-packing-system>



Components Used

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Servo Motor
- Motor Driver IC
- DC Motor
- Push Button
- LEDs
- Resistors

Working Principle

The ultrasonic sensor detects the object to be packaged. The Arduino processes the sensor input and controls the servo motor and DC motor through the motor driver circuit. The packaging mechanism operates automatically based on programmed automation logic, reducing manual effort and improving operational efficiency.

CONCLUSION

The internship in Robotics and Automation provided practical knowledge in automation systems, circuit designing, and sensor-based control applications. During the internship, three automation projects were successfully designed and simulated: Smart Home Lighting System, Conveyor Belt with Sensors, and Automated Packaging System.

These projects helped in understanding the working principles of sensors, microcontrollers, motors, and automation logic used in modern industrial and smart systems. The internship also improved technical skills in circuit design, troubleshooting, simulation, and embedded system applications.

The circuits were designed and tested using [Tinkercad](#), which provided a practical platform for implementing and analyzing automation concepts. Overall, the internship enhanced both technical knowledge and practical exposure in the field of Robotics and Automation.